



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal TO VALLI DEVAN FUELS

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## **Compliance with the Road Transport Safety Policy (RTSP)**

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## **Emergency Preparedness and Response**

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## **Environmental Considerations**

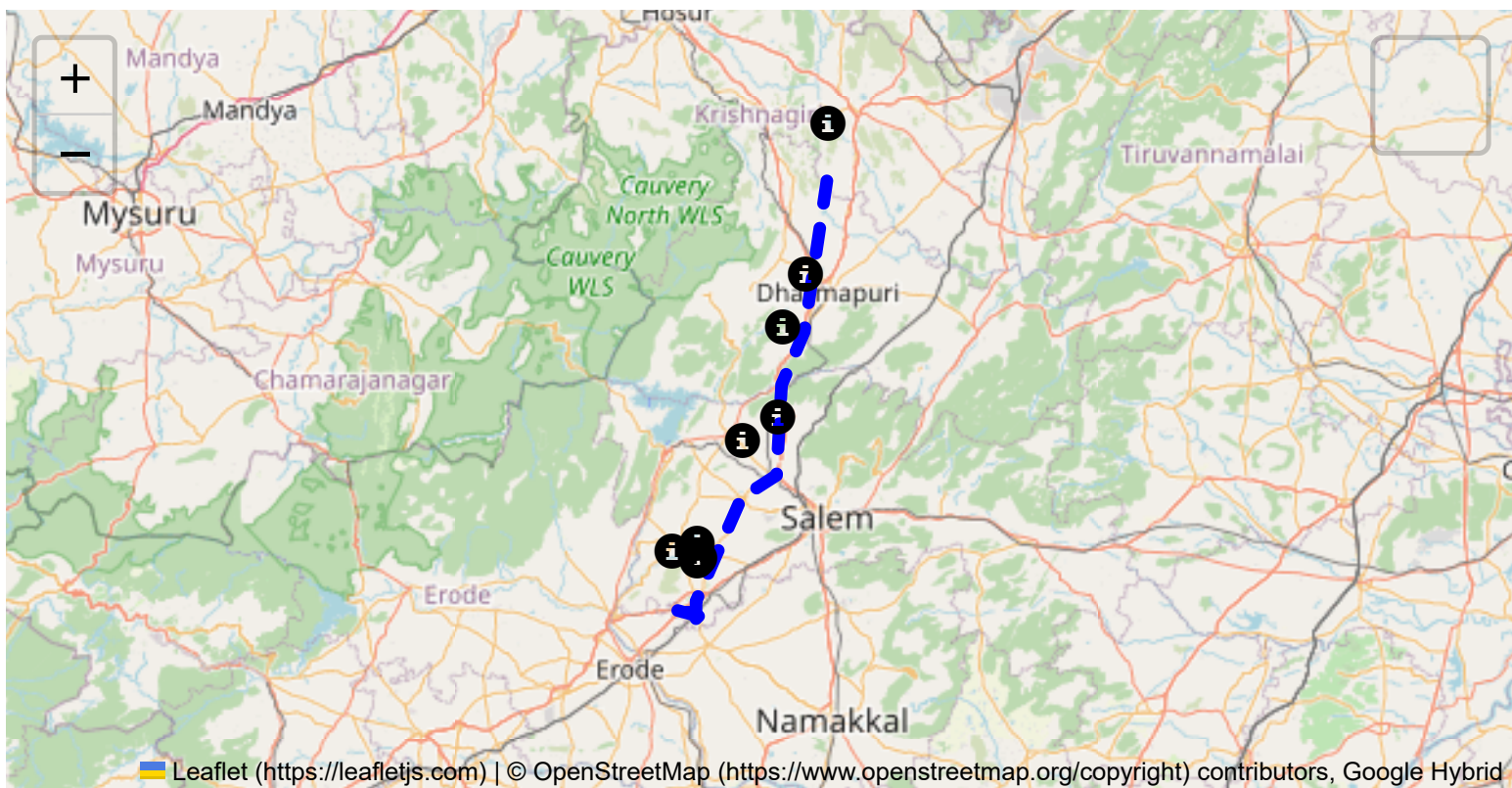
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

## **Journey Risk Mitigation**

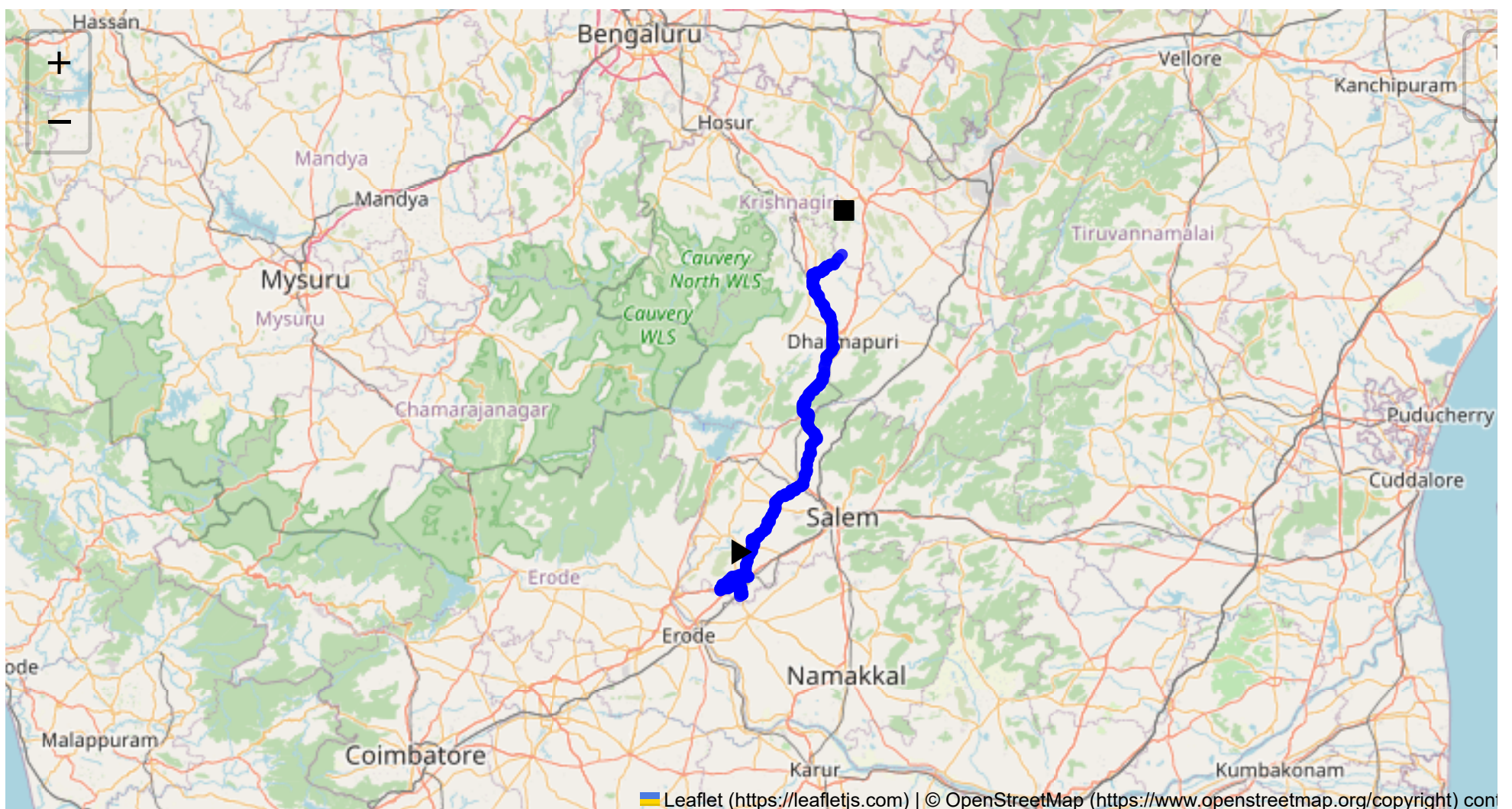
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

## **Compliance with Government Directives**

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



**Route Summary:**  
**Total Distance: 143.25 km**  
**Estimated Duration: 3.0 hours**  
**Adjusted Duration (Heavy Vehicle): 3.7 hours**  
**Start: (11.4381, 77.8734)**  
**End: (12.37305, 78.156916)**



## Welcome to the Journey Risk Management Study

### 1. Overview of the Route Map

The route from CVQF+23W, Sangagiri to 95F4+6Q Bannihalli involves travel through several towns and larger areas such as Kangeyampalayam, Padaiveedu, Salem, Tharamangalam, Omalur, Thoppur, and Seshampatti. The journey spans approximately 143.25 kilometers, primarily using major highways and

connecting roads. The route provides access to commercial and rural areas, requiring the navigation of diverse driving environments.

## 2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu typically experiences a tropical climate with dry winters, hot summers, and monsoons from June to September. Rain during monsoon can lead to waterlogged roads and reduced visibility. Drivers should be cautious of these conditions, especially in low-lying areas prone to flooding.

## 3. Analysis of Traffic Patterns

- **Peak Hours:** Typically, peak traffic hours coincide with the morning (7:30 AM - 10:00 AM) and evening (5:00 PM - 8:00 PM) commuter rush. Areas around Salem and other major urban points tend to be more congested.
- **Congestion-Prone Areas:** Salem, being a commercial hub, can experience significant traffic congestion. Construction or roadworks near Thoppur could also cause potential delays.

## 4. Assessment of Road Quality and Infrastructure

- The roads vary from well-maintained highways to narrower, less maintained connecting roads. The SH 86 and Bhavani Main Road are generally in good condition but can have potholes or uneven surfaces in rural stretches.
- Road markings and signage might be less frequent or visible in rural areas, requiring additional caution.

## 5. Suggestions for Alternative Routes for Emergencies

- **Alternative Route 1:** In severe congestion or roadblock scenarios in Salem, consider detouring via local roads parallel to the highway.
- **Alternative Route 2:** Utilizing state highways alongside NH44 near Thoppur for smoother transit, depending on real-time traffic data.

## 6. Summary of Local Regulations Affecting Hazardous Material Transport

Tamil Nadu regulations require permits for transporting hazardous materials, compliance with safety standards, and clearly displayed hazard placards. Time restrictions on heavy vehicles may vary, requiring attention to local signs or enforcement checkpoints.

## 7. Overview of Historical Incidents

- Limited data on historical incidents specific to hazardous materials in this exact route. However, Salem has witnessed a few heavy vehicle accidents due to speeding and improper vehicle maintenance, particularly in traffic-dense areas.

- Awareness of past incidents encourages vigilance regarding vehicle maintenance and adherence to speed limits.

## 8. Environmental Considerations and Sensitive Areas

- Rural areas en route are ecologically sensitive, particularly around agricultural land. Efforts should be made to minimize environmental impact through controlled driving (avoiding excessive idling and emissions).
- Watch for regional wildlife crossings in rural stretches, especially during dusk or dawn.

## 9. Analysis of Communication Coverage

- Generally, communication coverage is good along the major segments, such as highways.
- Potential dead zones are more likely in rural and remote stretches, especially around forested or hilly terrains.

## 10. Estimated Emergency Response Times

- **Salem and Thoppur:** As these are larger urban centers, emergency response times may be as quick as 15-30 minutes.
- **Rural segments:** Emergency services could take up to 45-60 minutes or more, depending on road conditions and traffic.

## 11. An Overall Summary of Risk Assessment

This route encompasses major highways with good infrastructure and certain rural roads requiring caution. The primary risks include weather-related conditions during monsoons, potential traffic congestion, and road quality challenges in less urbanized stretches. Regular communication checks and compliance with local regulations are critical for safe transit. Emergency preparedness measures should be a priority due to variable response times.

In summary, while this route is relatively straightforward, vigilance and preparation for variable traffic and weather conditions, along with observance of safety protocols, will be essential for ensuring safe passage of heavy vehicles carrying hazardous materials.

### Risk Assessment - Turns

|   | Risk Type | Risk Level | Coordinates           | Speed Limit |
|---|-----------|------------|-----------------------|-------------|
| 0 | U-Turn    | High       | 11.4749755, 77.869149 | 10 KM/Hr    |
| 1 | Turn      | High       | 11.43968, 77.87345    | 15 KM/Hr    |
| 2 | Turn      | High       | 11.44028, 77.87541    | 15 KM/Hr    |

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 3  | Turn       | High       | 11.44898, 77.87410 | 15 KM/Hr    |
| 4  | Turn       | High       | 11.45007, 77.87201 | 15 KM/Hr    |
| 5  | Turn       | Medium     | 11.46392, 77.87328 | 30 KM/Hr    |
| 6  | Turn       | High       | 11.47391, 77.86844 | 15 KM/Hr    |
| 7  | Blind Spot | Blind Spot | 11.47481, 77.86920 | 10 KM/Hr    |
| 8  | Turn       | Medium     | 11.47354, 77.86775 | 30 KM/Hr    |
| 9  | Turn       | Medium     | 11.45543, 77.81780 | 30 KM/Hr    |
| 10 | Turn       | Medium     | 11.45580, 77.81697 | 30 KM/Hr    |
| 11 | Turn       | High       | 11.45615, 77.81677 | 15 KM/Hr    |
| 12 | Blind Spot | Blind Spot | 11.49225, 77.89560 | 10 KM/Hr    |
| 13 | Turn       | Medium     | 11.49377, 77.88573 | 30 KM/Hr    |
| 14 | Turn       | High       | 11.49469, 77.88488 | 15 KM/Hr    |
| 15 | Turn       | High       | 11.55618, 77.89789 | 15 KM/Hr    |
| 16 | Turn       | Medium     | 11.55629, 77.89888 | 30 KM/Hr    |
| 17 | Turn       | High       | 11.58228, 77.90841 | 15 KM/Hr    |
| 18 | Turn       | High       | 11.58340, 77.90637 | 15 KM/Hr    |
| 19 | Turn       | Medium     | 11.59201, 77.90649 | 30 KM/Hr    |
| 20 | Turn       | Medium     | 11.63882, 77.94859 | 30 KM/Hr    |
| 21 | Turn       | Medium     | 11.64161, 77.95177 | 30 KM/Hr    |
| 22 | Turn       | Medium     | 11.64226, 77.95200 | 30 KM/Hr    |
| 23 | Turn       | Medium     | 11.64716, 77.95118 | 30 KM/Hr    |
| 24 | Turn       | Medium     | 11.69553, 77.97110 | 30 KM/Hr    |
| 25 | Turn       | Medium     | 11.69806, 77.97537 | 30 KM/Hr    |
| 26 | Turn       | Medium     | 11.71386, 77.99552 | 30 KM/Hr    |
| 27 | Turn       | Medium     | 11.71735, 78.00019 | 30 KM/Hr    |
| 28 | Turn       | Medium     | 11.71915, 78.01106 | 30 KM/Hr    |
| 29 | Turn       | Medium     | 11.72492, 78.01660 | 30 KM/Hr    |
| 30 | Turn       | Medium     | 11.72765, 78.02291 | 30 KM/Hr    |
| 31 | Turn       | Medium     | 11.73786, 78.03505 | 30 KM/Hr    |
| 32 | Turn       | High       | 11.73652, 78.03866 | 15 KM/Hr    |
| 33 | Turn       | Medium     | 11.74255, 78.04707 | 30 KM/Hr    |
| 34 | Turn       | Medium     | 11.77984, 78.05782 | 30 KM/Hr    |
| 35 | Turn       | Medium     | 11.86902, 78.08380 | 30 KM/Hr    |
| 36 | Turn       | Medium     | 11.87988, 78.08176 | 30 KM/Hr    |

|    | Risk Type | Risk Level | Coordinates        | Speed Limit |
|----|-----------|------------|--------------------|-------------|
| 37 | Turn      | Medium     | 11.92637, 78.06515 | 30 KM/Hr    |
| 38 | Turn      | Medium     | 11.93726, 78.05359 | 30 KM/Hr    |
| 39 | Turn      | Medium     | 11.95482, 78.04687 | 30 KM/Hr    |
| 40 | Turn      | High       | 11.95837, 78.04986 | 15 KM/Hr    |
| 41 | Turn      | High       | 11.96007, 78.04945 | 15 KM/Hr    |
| 42 | Turn      | Medium     | 11.96347, 78.05142 | 30 KM/Hr    |
| 43 | Turn      | High       | 11.96415, 78.05126 | 15 KM/Hr    |
| 44 | Turn      | Medium     | 11.96519, 78.04958 | 30 KM/Hr    |
| 45 | Turn      | High       | 11.96620, 78.04920 | 15 KM/Hr    |
| 46 | Turn      | Medium     | 11.96842, 78.05097 | 30 KM/Hr    |
| 47 | Turn      | Medium     | 12.12010, 78.13282 | 30 KM/Hr    |
| 48 | Turn      | Medium     | 12.26535, 78.08429 | 30 KM/Hr    |
| 49 | Turn      | Medium     | 12.27235, 78.08086 | 30 KM/Hr    |
| 50 | Turn      | High       | 12.29602, 78.07393 | 15 KM/Hr    |
| 51 | Turn      | High       | 12.29655, 78.07551 | 15 KM/Hr    |
| 52 | Turn      | High       | 12.31207, 78.07838 | 15 KM/Hr    |
| 53 | Turn      | High       | 12.31355, 78.07485 | 15 KM/Hr    |
| 54 | Turn      | Medium     | 12.31427, 78.07473 | 30 KM/Hr    |
| 55 | Turn      | High       | 12.32003, 78.07652 | 15 KM/Hr    |
| 56 | Turn      | Medium     | 12.32051, 78.08341 | 30 KM/Hr    |
| 57 | Turn      | High       | 12.32340, 78.08687 | 15 KM/Hr    |
| 58 | Turn      | Medium     | 12.32338, 78.09546 | 30 KM/Hr    |
| 59 | Turn      | Medium     | 12.33674, 78.10807 | 30 KM/Hr    |

## Emergency Locations

|   | type     | name                    | coordinates            | speed_limit | risk_level |
|---|----------|-------------------------|------------------------|-------------|------------|
| 0 | hospital | Thilagam Hospital       | 11.4753573, 77.8685829 | 30 km/h     | Medium     |
| 1 | clinic   | Meyyazhagan Clinic      | 11.4752254, 77.8685671 | 30 km/h     | Medium     |
| 2 | hospital | Dr. Jaganathan Hospital | 11.4752717, 77.8709312 | 30 km/h     | Medium     |
| 3 | hospital | Kathir Eye Hospital     | 11.4740976, 77.8681591 | 30 km/h     | Medium     |
| 4 | police   | Police Quarters         | 11.4741243, 77.8673495 | 30 km/h     | Medium     |

|    | type     | name                               | coordinates            | speed_limit | risk_level |
|----|----------|------------------------------------|------------------------|-------------|------------|
| 6  | hospital | Ck Hospital                        | 11.473504, 77.8650523  | 30 km/h     | Medium     |
| 7  | hospital | SP Hospital                        | 11.4708567, 77.8654872 | 30 km/h     | Medium     |
| 8  | hospital | Lakshmi Hospital                   | 11.4723746, 77.8646891 | 30 km/h     | Medium     |
| 9  | hospital | ESI Hospital                       | 11.4717067, 77.8642668 | 30 km/h     | Medium     |
| 10 | clinic   | Sutha Clinic                       | 11.473718, 77.862614   | 30 km/h     | Medium     |
| 12 | clinic   | Sun Eye Care                       | 11.6936242, 77.9730948 | 30 km/h     | Medium     |
| 13 | hospital | Dr. Pandiyarajan Hospital          | 11.692704, 77.969344   | 30 km/h     | Medium     |
| 14 | hospital | Dr. Sengodan Hospital              | 11.6959527, 77.9695325 | 30 km/h     | Medium     |
| 15 | hospital | Omalur Government Hospital         | 11.74161, 78.044296    | 30 km/h     | Medium     |
| 16 | clinic   | R.R. Nursing Home                  | 11.7394129, 78.0439861 | 30 km/h     | Medium     |
| 17 | hospital | Sri Hari Hospital                  | 11.7396726, 78.0440701 | 30 km/h     | Medium     |
| 18 | hospital | Sugumar Hospital                   | 11.7396031, 78.0426955 | 30 km/h     | Medium     |
| 19 | hospital | Sri Sugam Hospital                 | 11.7429821, 78.0461039 | 30 km/h     | Medium     |
| 20 | hospital | Saravana Hospital                  | 11.7409303, 78.0458308 | 30 km/h     | Medium     |
| 21 | hospital | Arun Hospital                      | 11.7429892, 78.0466881 | 30 km/h     | Medium     |
| 22 | clinic   | Saratha Clinic                     | 11.7422818, 78.0458175 | 30 km/h     | Medium     |
| 24 | clinic   | Sri Meenakshi Nursing Home, Salem  | 11.7449162, 78.0474637 | 30 km/h     | Medium     |
| 25 | clinic   | Divyam Child Care Clinic           | 11.7447135, 78.0473858 | 30 km/h     | Medium     |
| 26 | clinic   | Momota Clinic                      | 11.744982, 78.047397   | 30 km/h     | Medium     |
| 27 | clinic   | Gomathi Clinic                     | 11.7445293, 78.0470401 | 30 km/h     | Medium     |
| 28 | hospital | Vinayaka Child Hospital            | 11.745812, 78.0480225  | 30 km/h     | Medium     |
| 29 | hospital | Dr. G.R.Thilakam Hospital          | 11.746097, 78.04769    | 30 km/h     | Medium     |
| 30 | hospital | Amirthalingam Hospital             | 11.7456708, 78.0470548 | 30 km/h     | Medium     |
| 31 | hospital | Children Hospital                  | 11.7458377, 78.0480205 | 30 km/h     | Medium     |
| 32 | clinic   | Shri Ramana Poly Clinic            | 11.7477122, 78.0479293 | 30 km/h     | Medium     |
| 33 | hospital | Dr. Nagaraj Hospital               | 11.7464035, 78.0472401 | 30 km/h     | Medium     |
| 34 | hospital | Thulasi Hospital                   | 11.7461516, 78.0469984 | 30 km/h     | Medium     |
| 35 | clinic   | Manimuthu Skin Clinic              | 11.7521189, 78.0484058 | 30 km/h     | Medium     |
| 36 | hospital | St. Joseph's Speciality Hospital   | 11.761611, 78.0522344  | 30 km/h     | Medium     |
| 38 | hospital | Anugraha Multi Speciality Hospital | 11.773549, 78.0546408  | 30 km/h     | Medium     |
| 40 | hospital | L.R. Hospital                      | 11.8703354, 78.0844554 | 30 km/h     | Medium     |
| 41 | clinic   | JSA Clinic                         | 11.8700619, 78.0844982 | 30 km/h     | Medium     |
| 42 | hospital | Thoppur,Government Hospital        | 11.9398824, 78.0551658 | 30 km/h     | Medium     |
| 43 | hospital | Government Hospital Palacode       | 12.2964824, 78.0761735 | 30 km/h     | Medium     |

## Crowded Spots

|    | type    | name                                              | coordinates               | speed_limit | risk_level |
|----|---------|---------------------------------------------------|---------------------------|-------------|------------|
| 5  | school  | Goverment Boys Higher Secondary School            | 11.477659,<br>77.8659632  | 30 km/h     | Medium     |
| 11 | school  | KRP Matric. Hr. Sec School                        | 11.4546193,<br>77.8142445 | 30 km/h     | Medium     |
| 23 | school  | Fathima Hr Sec School                             | 11.7410508,<br>78.0479272 | 30 km/h     | Medium     |
| 37 | school  | John Britto Matriculation Higher Secondary School | 11.7676907,<br>78.0513092 | 30 km/h     | Medium     |
| 39 | college | Dhirajlal Gandhi College of Technology            | 11.7968014,<br>78.0562095 | 30 km/h     | Medium     |

## Route Photos of Risky Spots



**Risk Type:** U-Turn

**Risk Level:** High

**Speed Limit:** 10 KM/Hr

**Coordinates:** 11.4749755, 77.869149



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.44028, 77.87541



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.44898, 77.87410



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.45007, 77.87201



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.46392, 77.87328



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.47391, 77.86844



**Risk Type:** Blind Spot

**Risk Level:** Blind Spot

**Speed Limit:** 10 KM/Hr

**Coordinates:** 11.47481, 77.86920



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.47354, 77.86775



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45543, 77.81780



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45580, 77.81697



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.45615, 77.81677



**Risk Type:** Blind Spot

**Risk Level:** Blind Spot

**Speed Limit:** 10 KM/Hr

**Coordinates:** 11.49225, 77.89560



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.49377, 77.88573



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.49469, 77.88488



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.55618, 77.89789



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.58340, 77.90637



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.59201, 77.90649



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.63882, 77.94859



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.64161, 77.95177



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.64226, 77.95200



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.64716, 77.95118



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.69553, 77.97110



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.69806, 77.97537



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.71386, 77.99552



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.71735, 78.00019



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.71915, 78.01106



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.72492, 78.01660



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.72765, 78.02291



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.73786, 78.03505



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.73652, 78.03866



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.74255, 78.04707



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.77984, 78.05782



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.86902, 78.08380



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.87988, 78.08176



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.92637, 78.06515



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.93726, 78.05359



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.95482, 78.04687



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.95837, 78.04986



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.96007, 78.04945



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.96347, 78.05142



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.96415, 78.05126



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.96519, 78.04958



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.96620, 78.04920



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.96842, 78.05097



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.12010, 78.13282



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.26535, 78.08429



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 12.27235, 78.08086



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 12.29602, 78.07393



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 12.29655, 78.07551



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 12.31207, 78.07838



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 12.31355, 78.07485



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 12.31427, 78.07473



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 12.32003, 78.07652



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 12.32051, 78.08341



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 12.32340, 78.08687



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 12.32338, 78.09546



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 12.33674, 78.10807

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